

Functional Specification Document (FSD)

Book a Doctor Platform

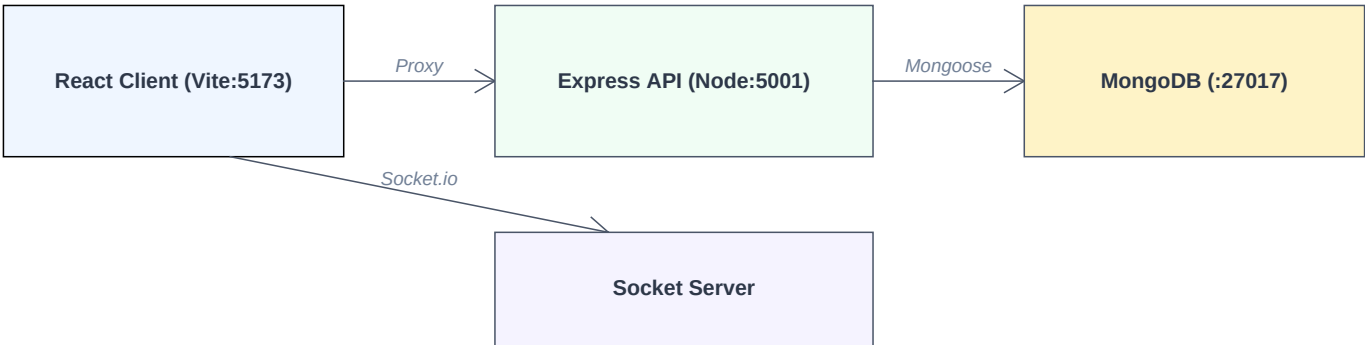
1. Executive Summary and Project Goal

The 'Book a Doctor' platform is a comprehensive full-stack healthcare scheduling web application built using the MERN stack (MongoDB, Express, React, Node.js). The platform is designed to digitize healthcare booking, remove queues, verify doctor credentials, and provide real-time updates for patients, doctors, and system administrators. The primary goal is to address scheduling inefficiencies, improve patient transparency, and automate clinic slot allocation.

2. Solution Architecture

The application operates as a Single Page Application (SPA) compiled under Vite v8 on the frontend and an Express REST and WebSocket server on the backend. Vite acts as a reverse proxy routing requests to the Express server, preventing CORS anomalies and network exposure complexities. Communications between the clients and the server occur over secure HTTP and real-time Socket.io channels.

Architecture Diagram



3. Database Schema Design (Mongoose Models)

- **User Model:** Maintains account types (Patient, Doctor, Admin), details, registration parameters, and double password hashing checks.
- **Doctor Model:** Tracks clinical details, experience levels, hourly slot prices, hospital affiliations, verification standing, and review ratings.
- **Appointment Model:** Saves time slots, status progressions (Pending, Approved, Completed, Cancelled), tele-links, and digital prescriptions.
- **Notification Model:** Stores notifications broadcasted over Socket.io.
- **Review Model:** Handles patient evaluations and stars ratings.

4. Feature Breakdown and User Journeys

- **Patient Dashboard:** Search and filter doctors by fee/rating, reserve slots on Interactive Calendar, check real-time statuses, and print prescriptions.
- **Doctor Dashboard:** Manage booking queue, toggle availability checkboxes, and write prescriptions.

- **Admin Dashboard:** Inspect analytics curves, verify practices, and register new doctors via add form.